

Insurance Campaign Response Dashboard

Test Case #1A · Who accepts the PA and Life insurance offers, what drives it, and how to target the next campaign · Source: dsc_test_case.csv

Segment

All

Acceptance rate

1.22%

2,621 of 214,493 customers bought · trend Jan→Dec



Customers contacted

214,493

offers across 12 campaign months

PA insurance

0.76%

1,633 policies sold

Life insurance

0.46%

988 policies sold

Product mix

62 / 38

PA / Life share of sales

Key findings

All customers · details in the sections below

1 in 82 customers accepts

Overall acceptance is **1.22%** (214k offers, 2.6k sales). PA sells more than Life (62% / 38% of sales). With 98.8% rejections, the target is heavily imbalanced.

Active accounts convert 3× better

Half of the customers had **no cash inflow** in the last 30 days, and they accept at 0.71%, against **2.07%** at 50k+ inflow. The gap is almost all PA. Customers 1–12 months on book reach 2.1%.

Younger customers respond best

Customers aged ≤25 accept at **2.0%**, falling to ~1.0% after 35. The drop is driven by **Life**: 0.72% at ≤25 vs 0.24% at 51+.

Product fit follows affluence

PA is the product for Lower Mass and incomes below 30k. **Life overtakes PA** in Upper Mass (0.68% vs 0.53%), incomes ≥30k and credit-card holders (0.65% vs 0.32%).

Smaller campaigns, better yield

Jun–Sep campaigns contacted about half as many customers but reached ~**1.9%** acceptance, vs 0.8–1.3% in the big months. More contacts did not bring proportionally more sales.

A model can halve the contact list

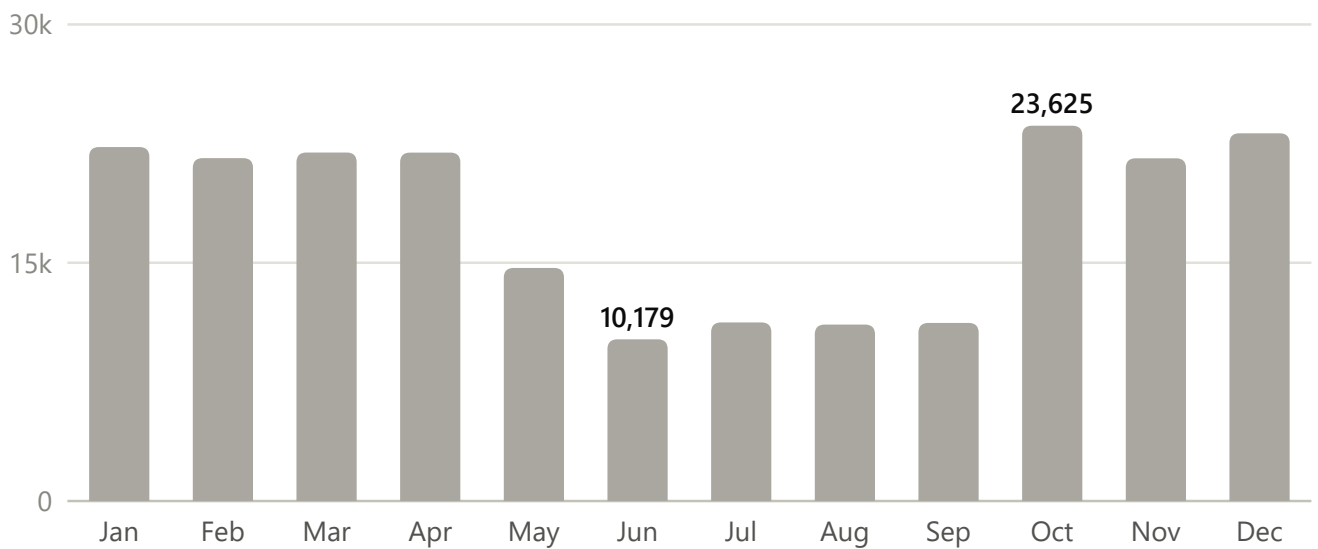
Contacting only the top 20% of customers ranked by the model would reach 49% of buyers. The top 10% buy at 3.6%, 3.0× the average (ROC-AUC 0.72).

1 • Campaign performance by month

Contact volume and acceptance rate · all customers

Customers contacted

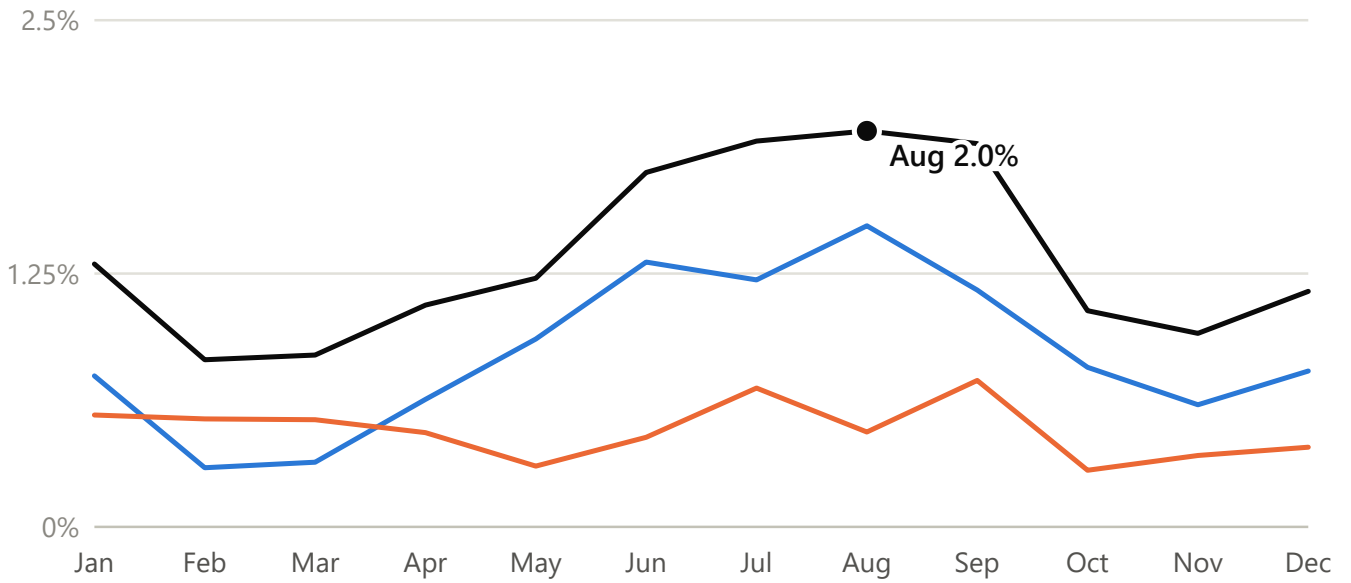
Offers per campaign month



Acceptance rate trend

% of contacted customers who bought · hover for each month

— Total — PA — Life



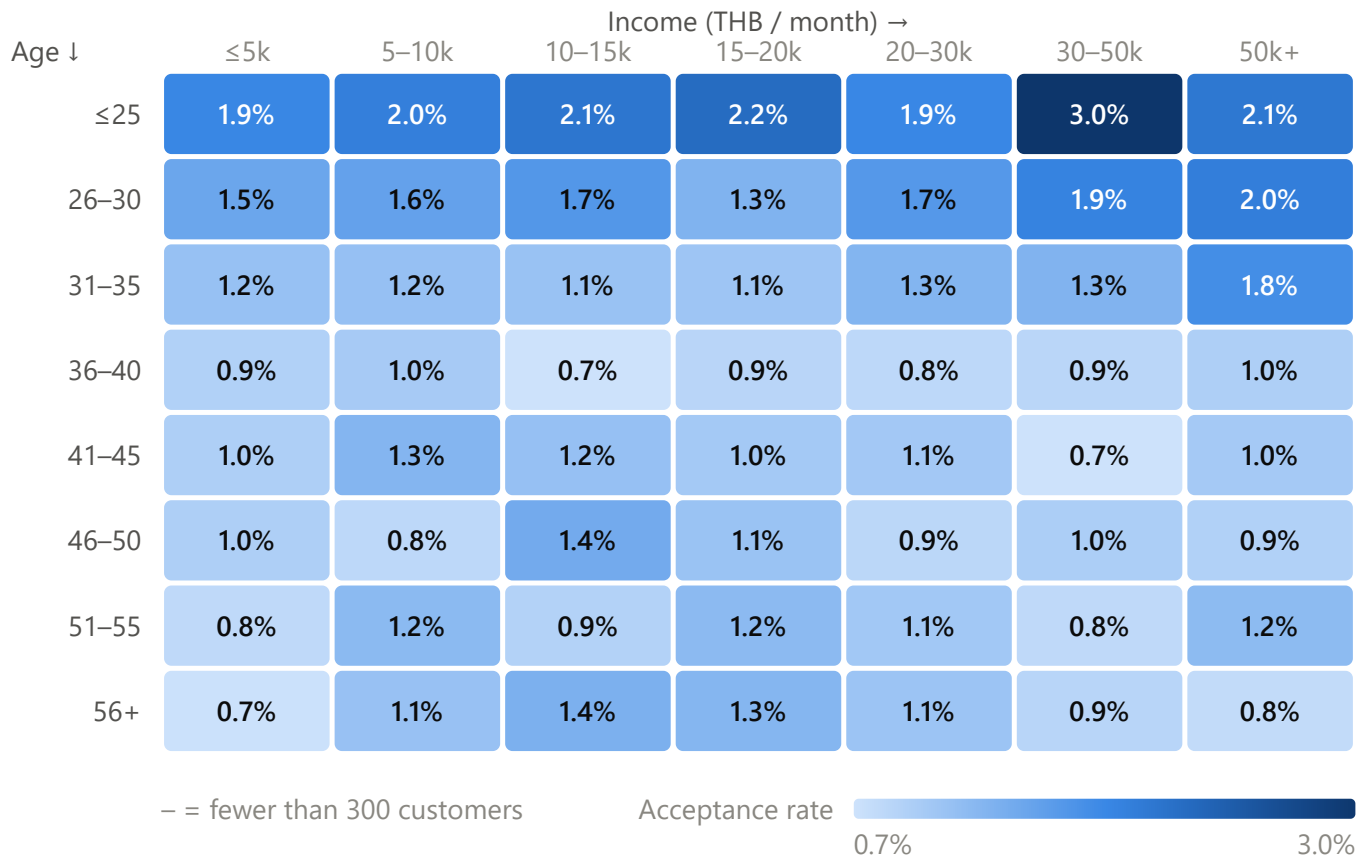
Acceptance roughly doubles in the low-volume months (Jun–Sep). The PA line drives the swing; Life stays between 0.3% and 0.7%.

2 · Where the buyers are

Combinations of customer attributes · all customers

Acceptance heatmap: age × income

Darker = higher acceptance rate

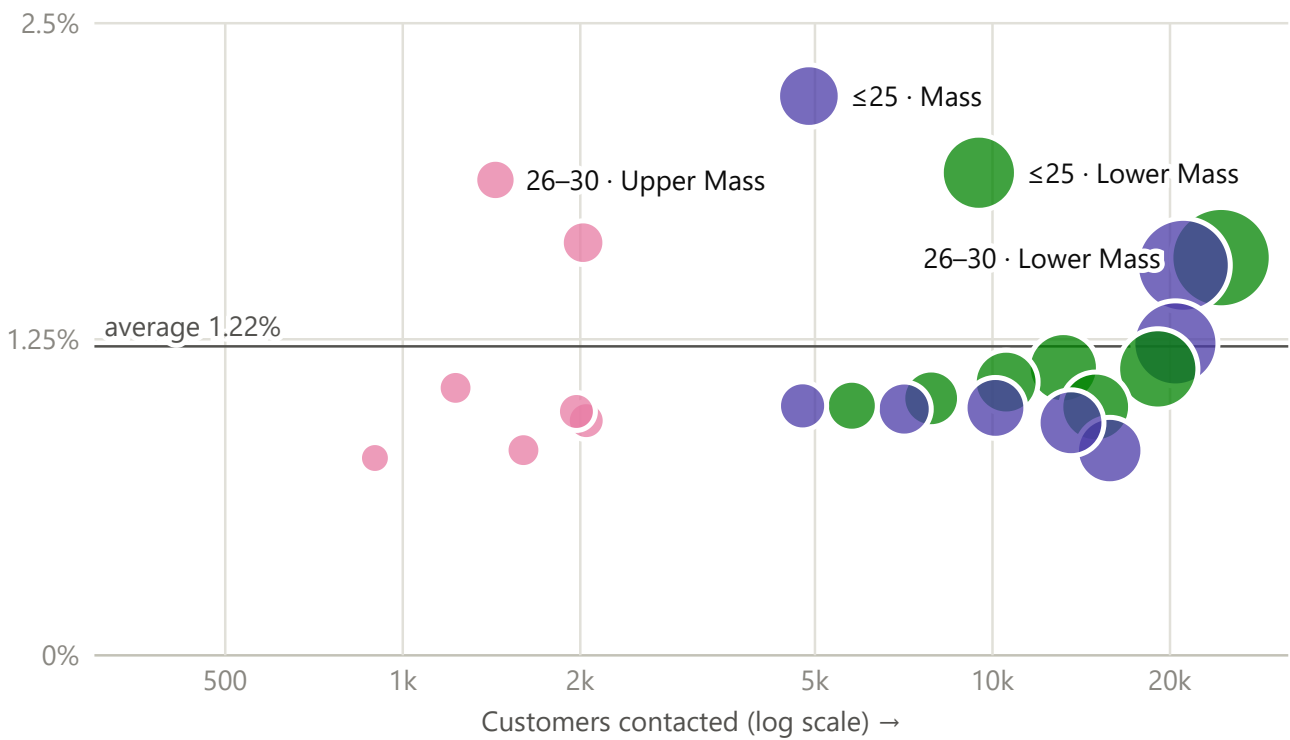


Young customers convert well at every income level. Above age 35, acceptance is low across the board.

Opportunity matrix: age band × segment

x = customers contacted · y = acceptance rate · bubble size = number of buyers

● Lower Mass ● Mass ● Upper Mass



Top-right bubbles are big groups that also convert well (young Lower Mass and Mass customers). Groups below the line take contact effort for little return.

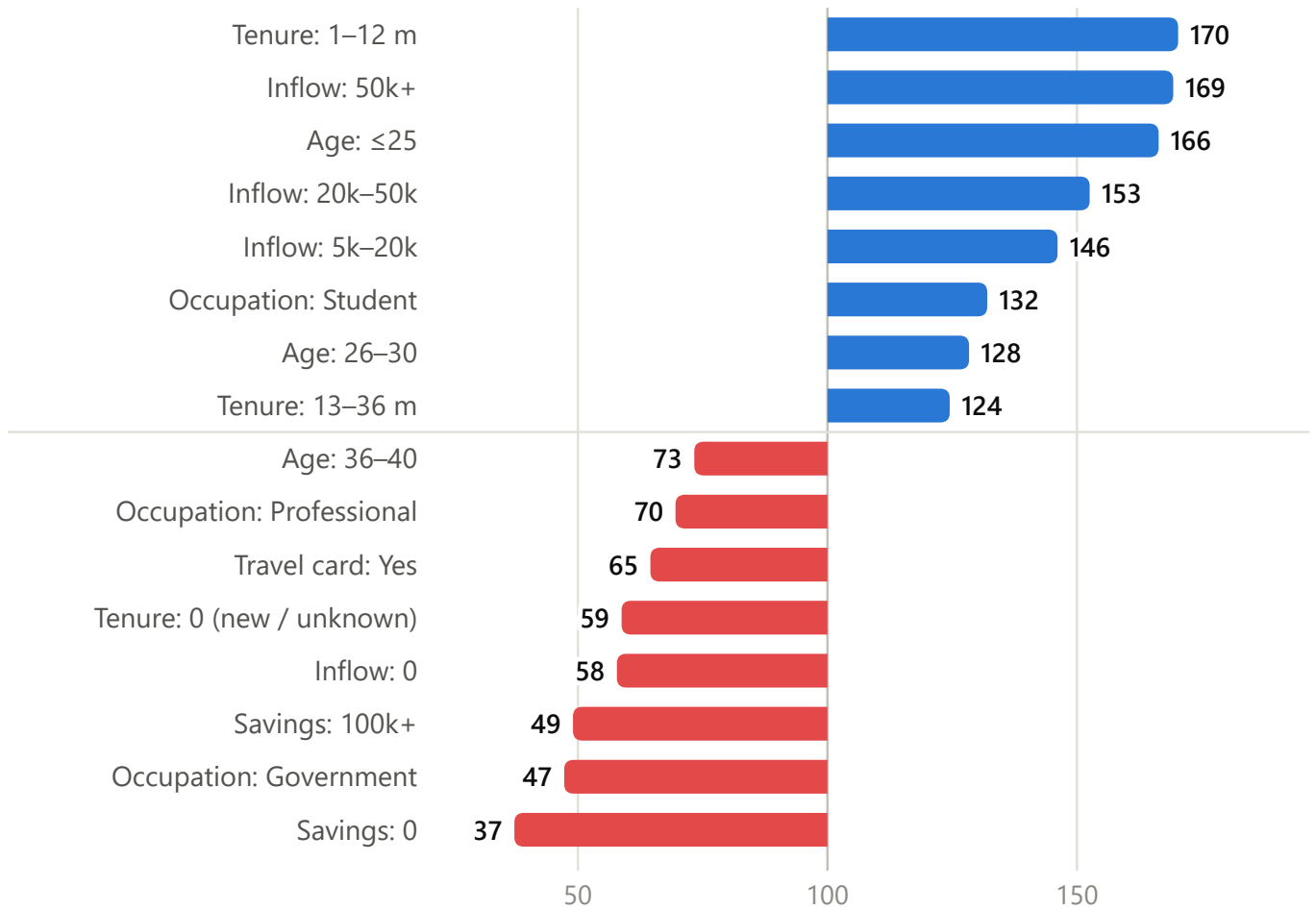
3 · What drives acceptance

How buyers differ from customers who rejected · all customers

Response index: best and worst groups

Group acceptance $\div$ average $\times 100 \cdot 100$ = average $\cdot$ groups with $\geq 1,000$ customers

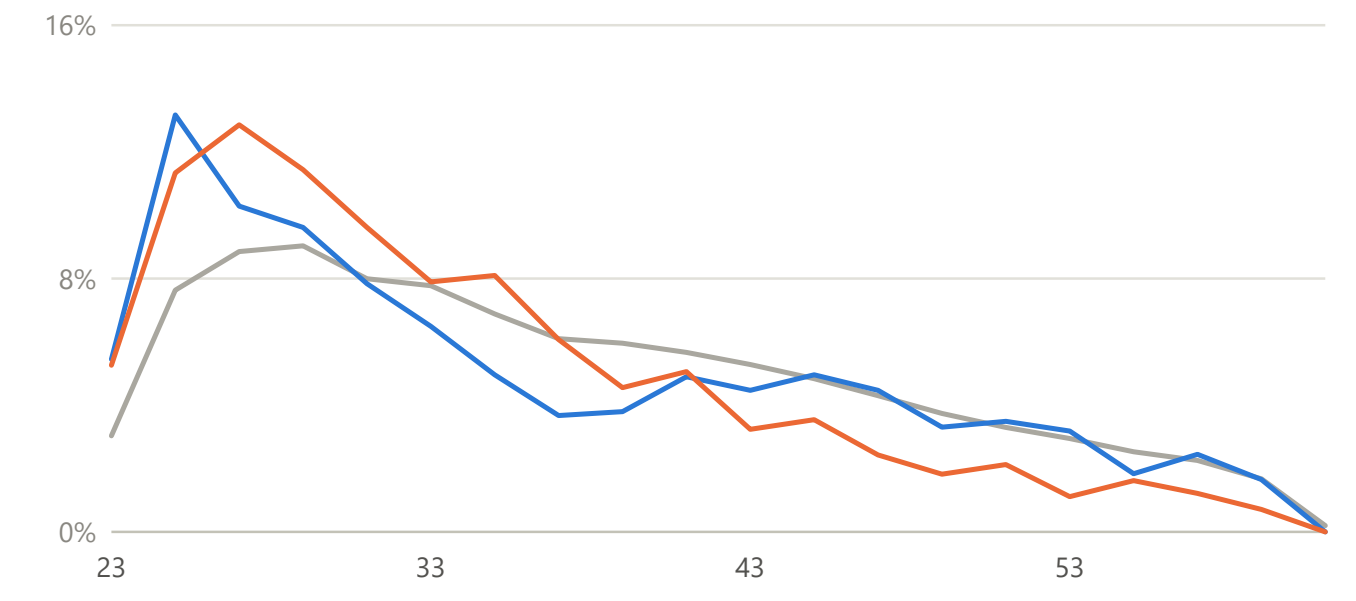
■ Above average ■ Below average



Age distribution by outcome

% of each group at each age (2-year bins)

Rejected PA buyers Life buyers

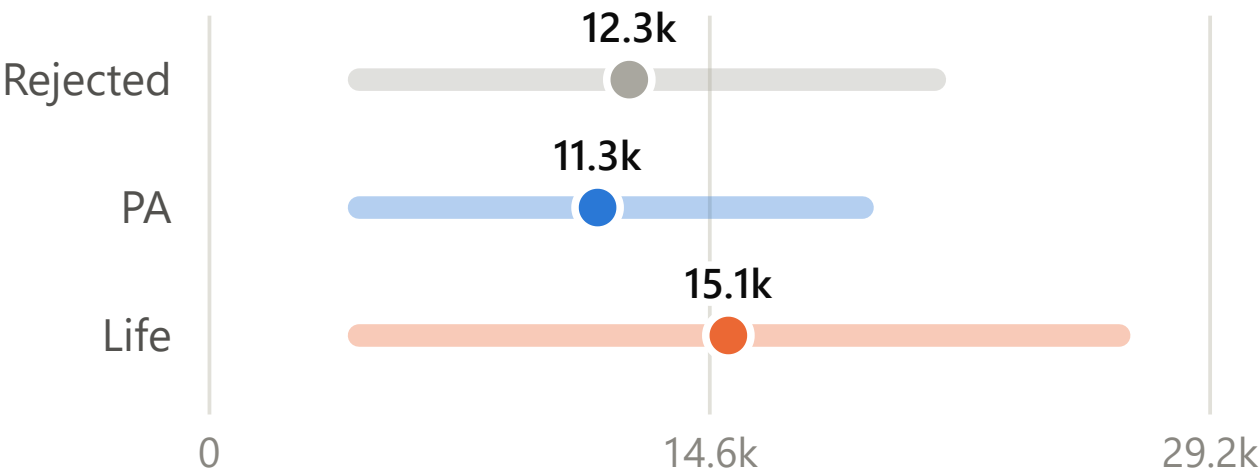


Life buyers peak in their mid-20s and PA buyers sit slightly younger than rejecters, so both buyer groups skew younger than the customer base.

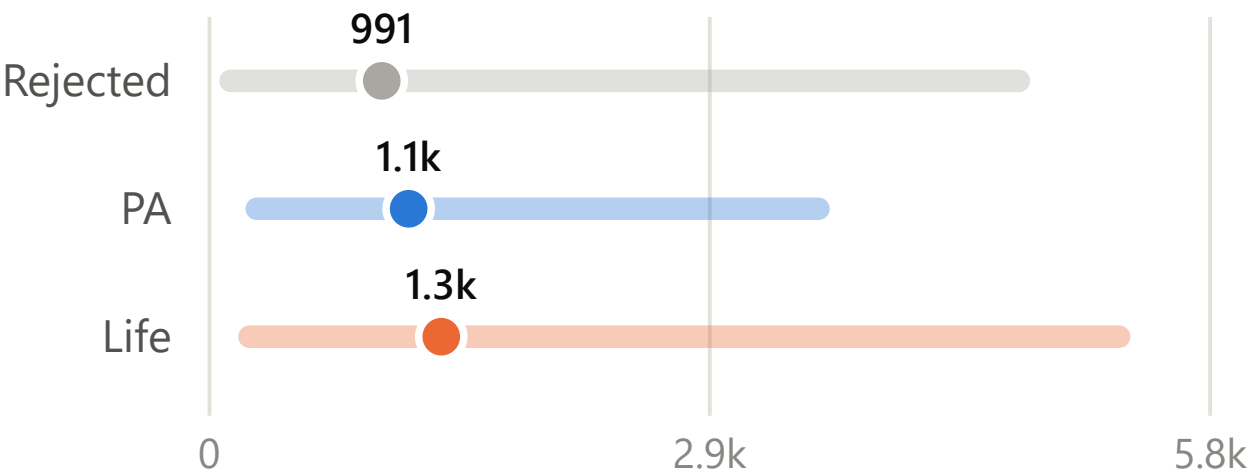
Median and middle 50% by outcome

Dot = median · band = 25th to 75th percentile (medians are used because the money fields are heavily skewed)

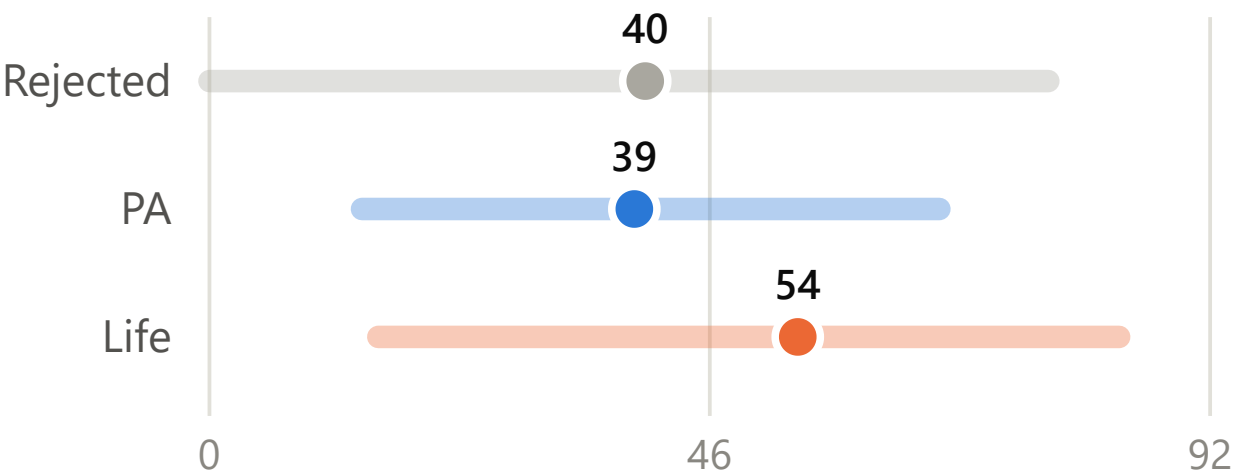
Monthly income (THB)



Avg. savings balance (THB)



Months on book

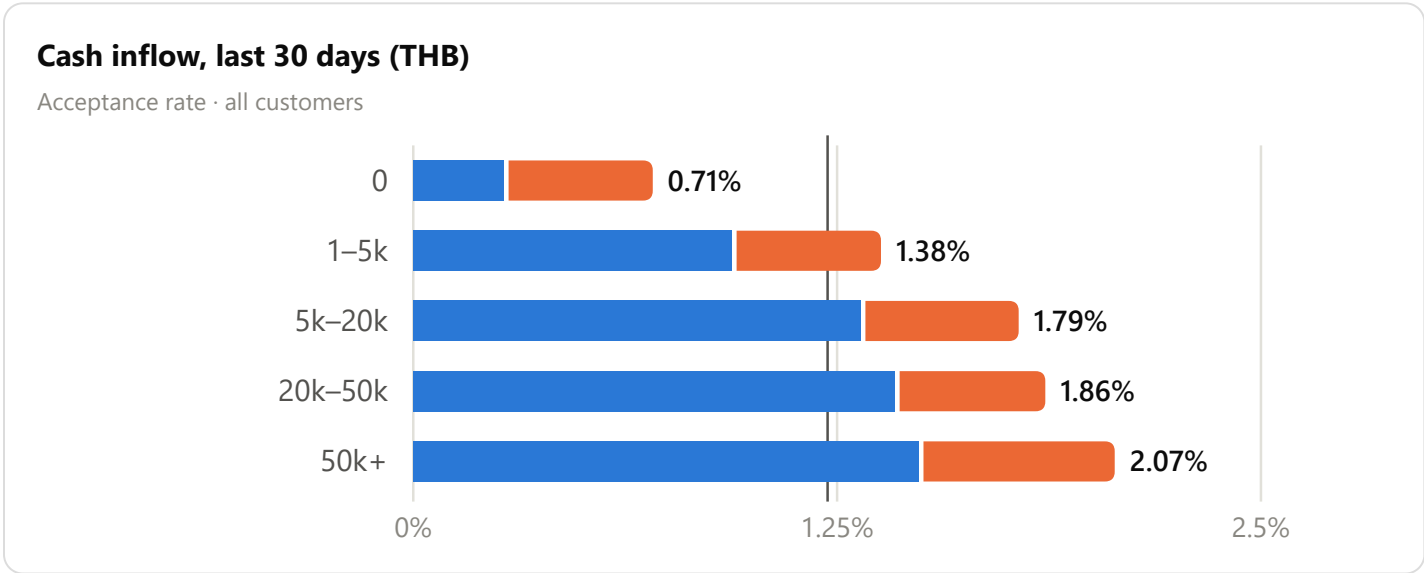
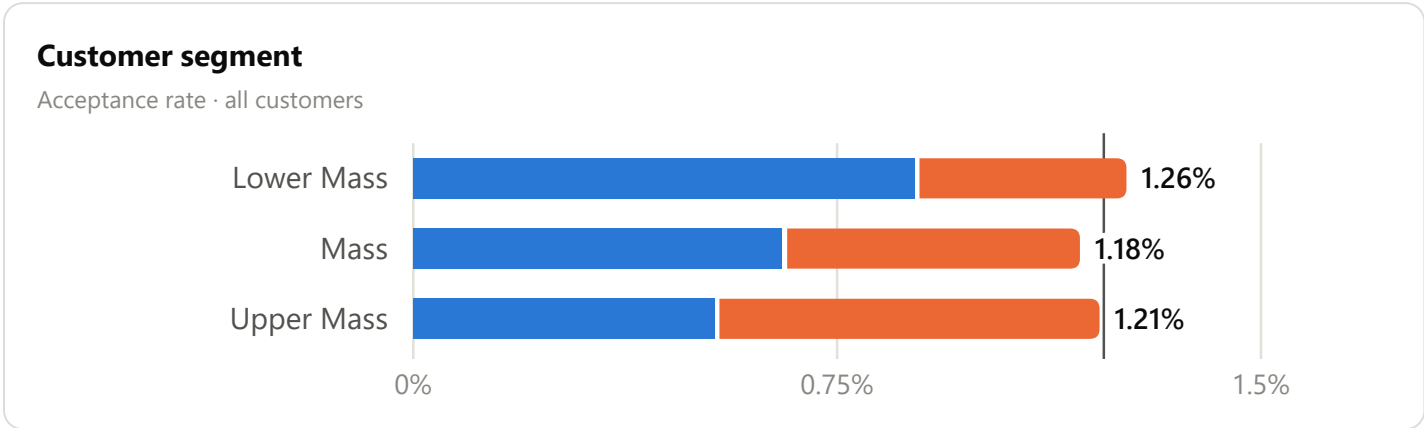


Life buyers earn more (median 15.1k vs 12.3k) and have longer tenure. PA buyers have lower income than rejecters but are more active.

4 · Acceptance by customer profile

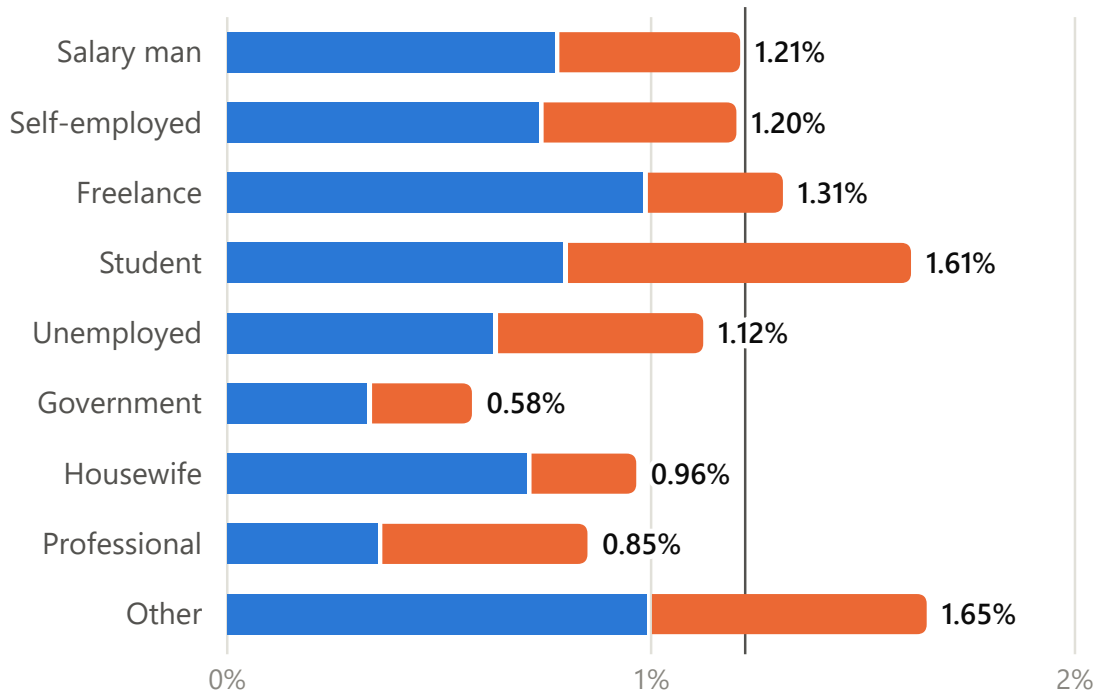
Bar length = total acceptance rate, split by product · vertical line = average · all customers

PA insurance Life insurance



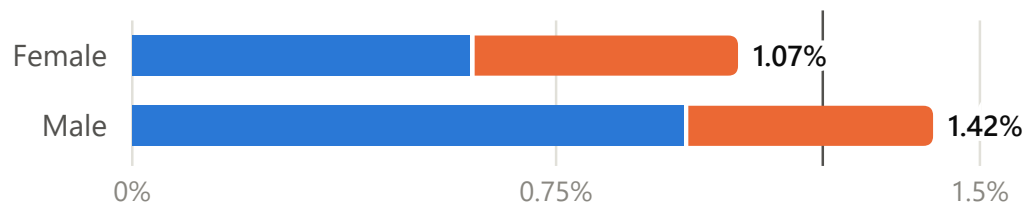
Occupation

Acceptance rate · all customers



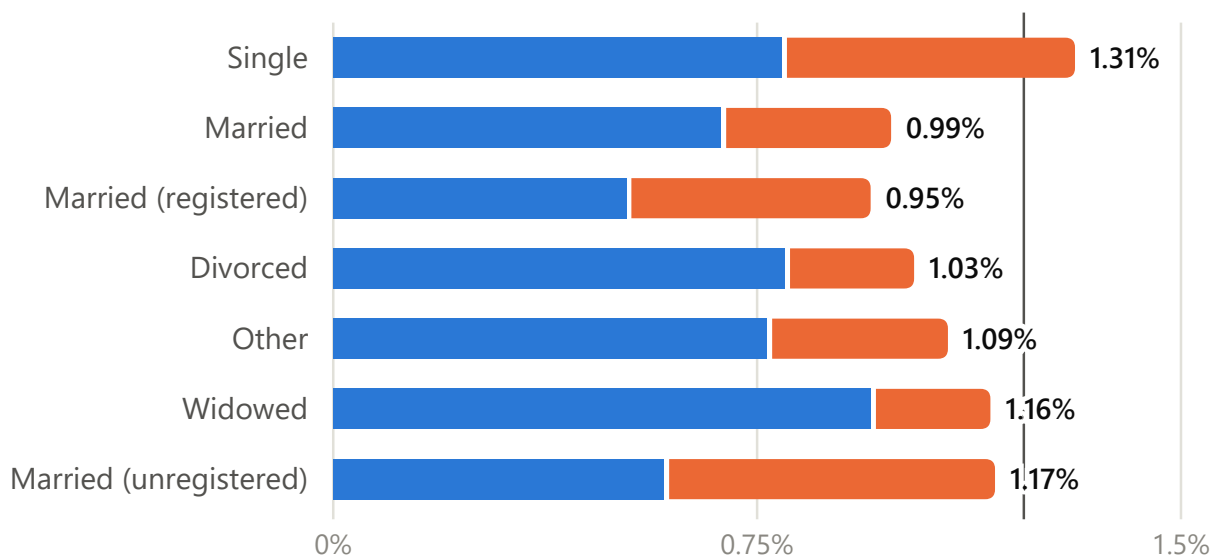
Gender

Acceptance rate · all customers



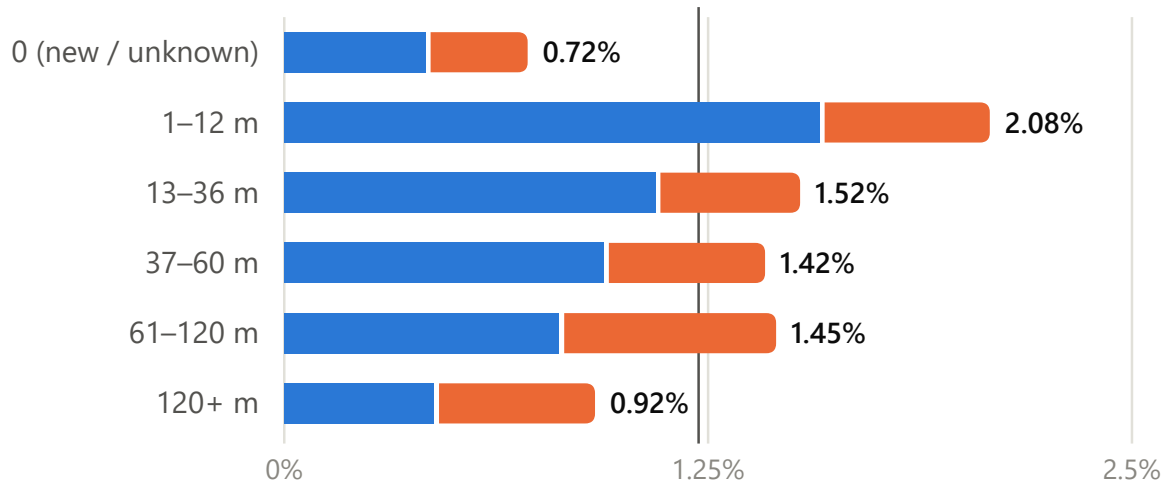
Marital status

Acceptance rate · all customers



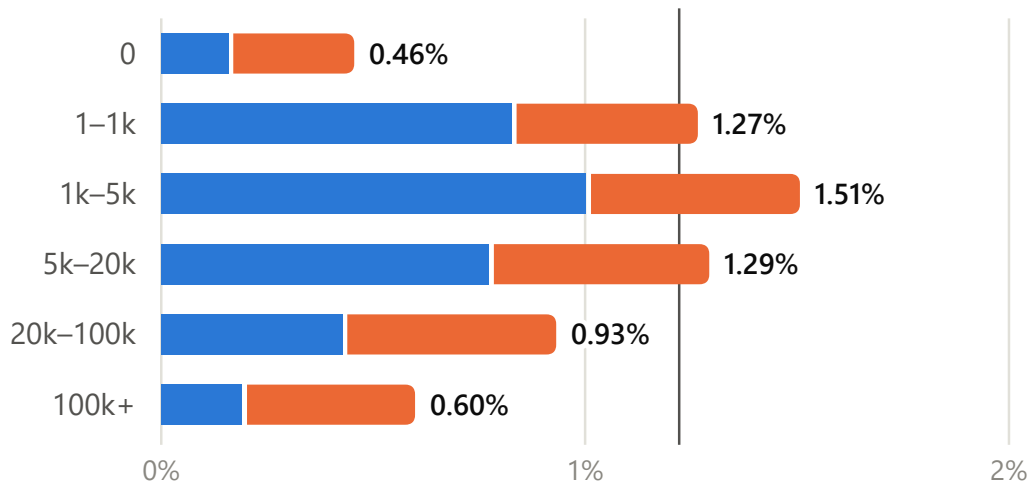
Months on book (tenure)

Acceptance rate · all customers



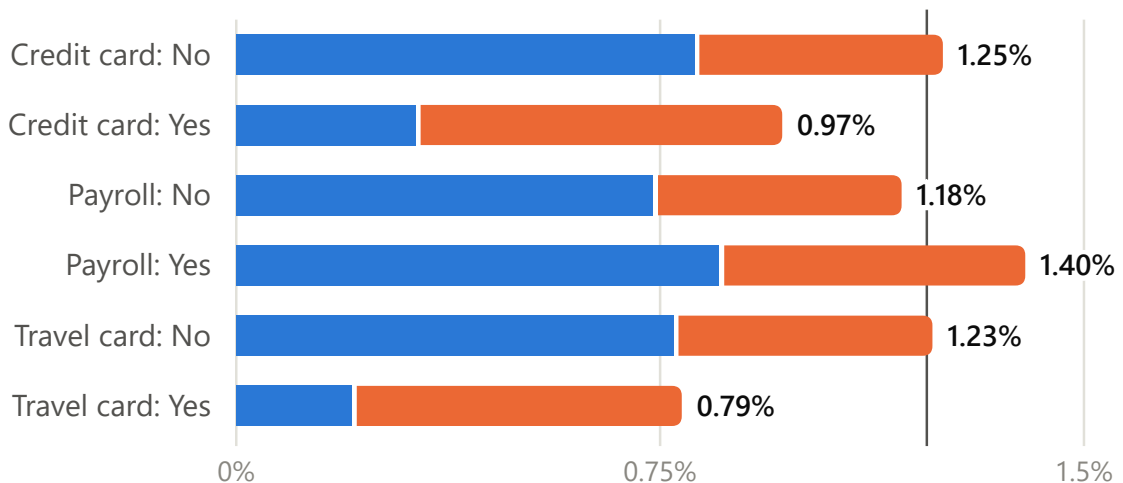
Avg. savings balance, 30 days (THB)

Acceptance rate · all customers



Product holding

Acceptance rate · all customers



Buyer profile comparison

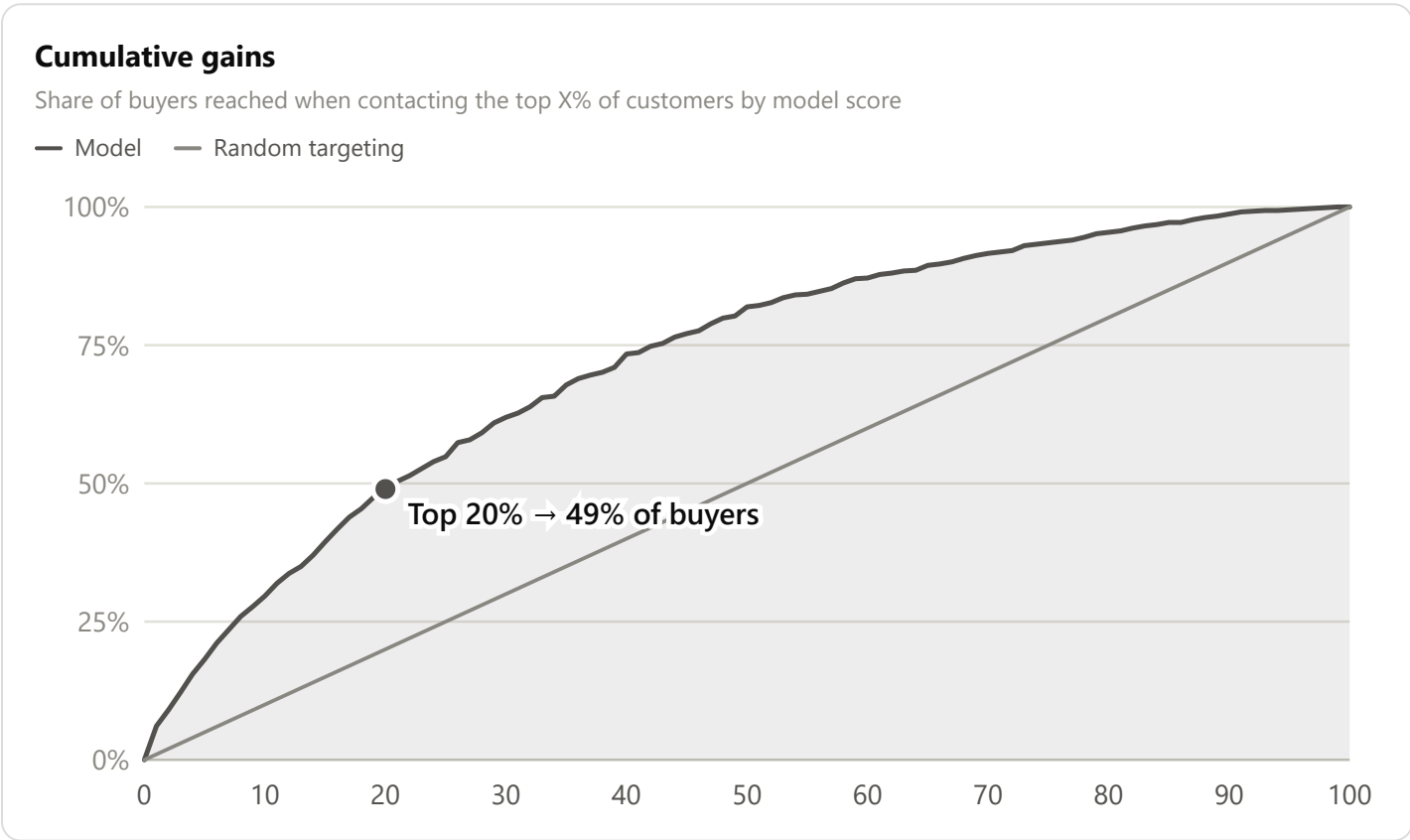
All customers

Measure	Rejected	Bought PA	Bought Life
Customers	211,872	1,633	988
Median age	36	34	32
Median income (THB)	12,250	11,325	15,140
Median avg. savings 30d (THB)	991	1,145	1,333
Median months on book	40	39	54
% male	43%	56%	42%
% Upper Mass	5.3%	3.7%	7.9%
% with credit card	10.0%	4.2%	14.1%
% on payroll	20.4%	22.9%	24.0%
% single	71%	74%	80%

5 · Propensity model: who to contact next time

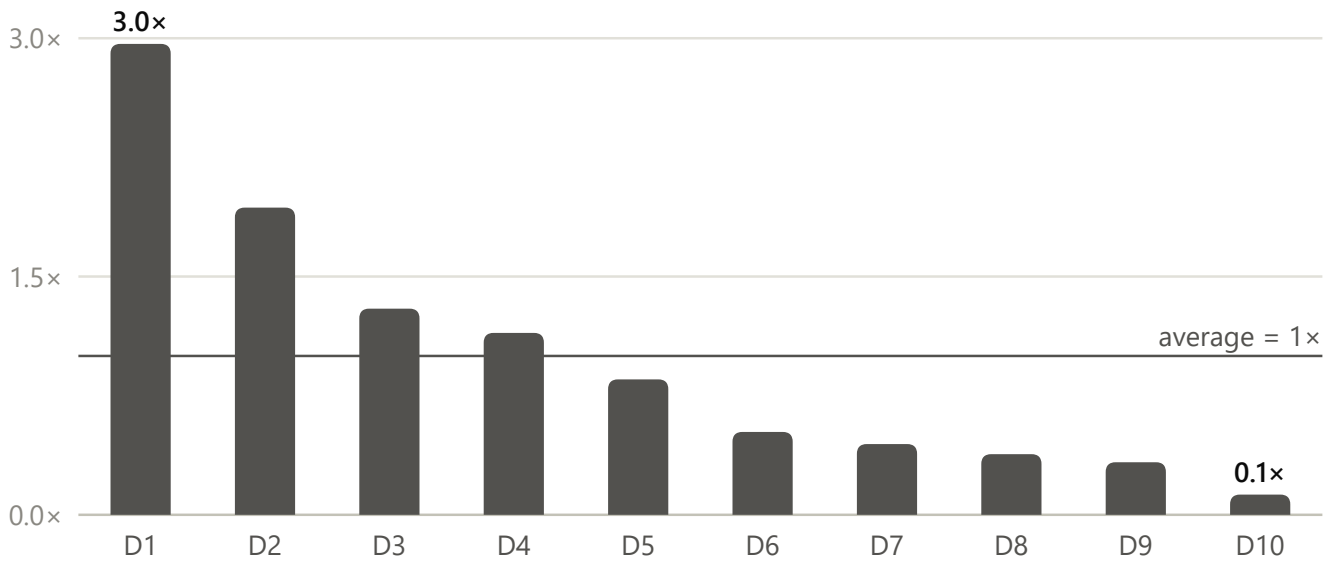
Gradient-boosted trees predicting "accepts any offer" · trained on 70%, evaluated on a 30% hold-out · all customers (not affected by the filter)

ROC-AUC 0.723 0.5 = random · 1.0 = perfect	Top 20% captures 49% of all buyers in the test set	Top-decile lift 3.0× 3.6% vs 1.2% average	PR-AUC 0.037 3.0× the 1.2% baseline
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Lift by score decile

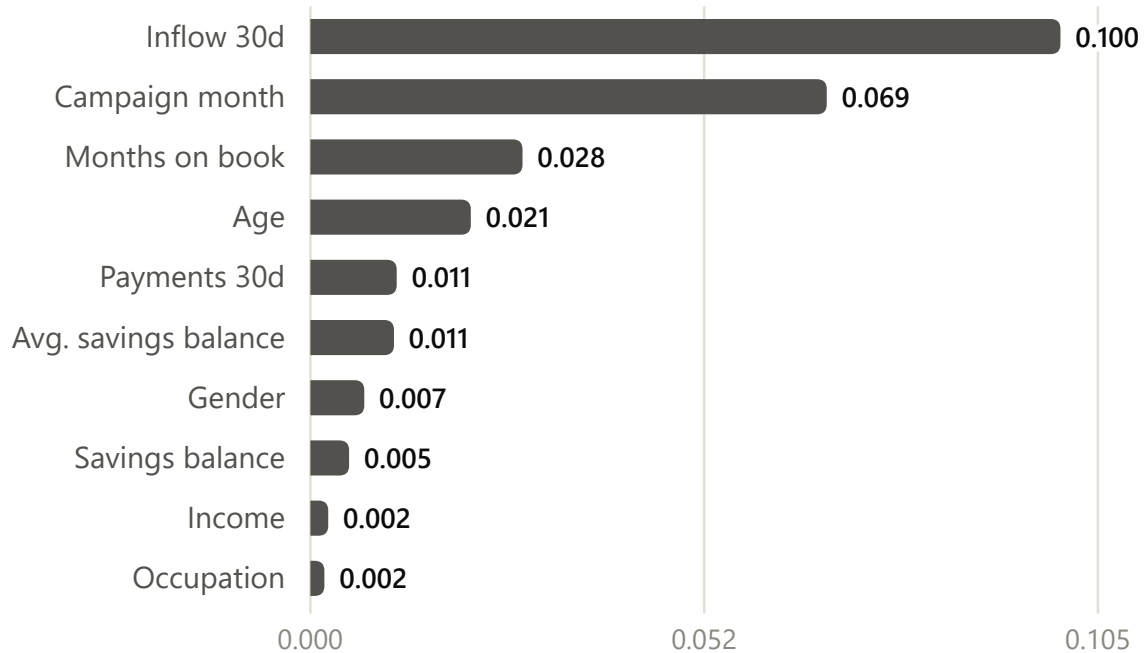
Acceptance in each 10% slice ÷ average · D1 = highest scores



The first four deciles beat the average, and D1 alone converts at 3×. Deciles 6–10 convert at about half the average or less, so those customers are candidates to skip.

What the model relies on

Permutation importance: how much ROC-AUC drops when the feature is shuffled



Cash inflow, campaign month, tenure and age carry most of the signal, which matches the descriptive analysis above.

Suggested contact tiers

Turn the score into an action: who gets which treatment

Tier	Customers	Acceptance	Lift	Share of buyers	Treatment
A · Call first	Top 0–20%	2.99%	2.4×	49%	Agent call, best offer
B · Standard	Top 20–50%	1.34%	1.1×	33%	Standard campaign
C · Low priority	Top 50–100%	0.44%	0.4×	18%	Digital only or skip

Tier A + B is 50% of the list but holds most of the buyers. Tier C can move to low-cost channels (SMS or app push) or be skipped.

Recommendations

1. **Score customers before every campaign.** Contacting the model's top 20–30% keeps about half to two-thirds of the sales at a fraction of the contact cost.
2. **Match the product to the customer.** Offer **Life** to Upper Mass, income $\geq 30k$, credit-card holders and the young; offer **PA** to active accounts with cash inflow, men, freelancers and incomes below 30k.
3. **Prioritise active, newer and younger customers:** 30-day inflow above 5k, 1–36 months on book, age ≤ 30 , savings 1k–20k.
4. **Contact fewer, better customers in the big months.** Jun–Sep converted at almost twice the rate with half the volume.
5. **Fix the data before the next model:** repair the 15-day flow fields, the tenure values and the empty rows, and add a customer ID plus the Policy and Payment data to track persistency after the sale.

Dataset summary

Rows 215,993 214,493 analysed after cleaning	Columns 26 25 features + label	Campaign months 12 Jan–Dec, no year given	Target (label) 3 classes 0 Reject · 1 PA · 2 Life
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Field	Type	Definition	Missing	Profile
campaign_month	string	Abbreviated Name of the Campaign Month	–	12 values · top: Oct (11%), Dec (11%), Jan (10%)
marital_sta	string	Marital Status	4,635 (2.15%)	8 values · top: Single (72%), Married (22%), Married (registered) (3%)
main_occupation	string	Main Occupation Class (might not really updated)	1,500 (0.69%)	10 values · top: Salary man (48%), Self-employed (26%), Freelance (9%)
customer_segment	string	Customer Segment - Lower Mass, Mass, Upper Mass	1,500 (0.69%)	3 values · top: Lower Mass (49%), Mass (46%), Upper Mass (5%)
gender	string	Customer Gender	1,500 (0.69%)	2 values · top: Female (57%), Male (43%)
have_acc_planet	string	Flag whether the customer has travel card account or not.	2,401 (1.11%)	2 values · top: N (98%), Y (2%)
have_cc	string	Flag whether the customer used to has Credit Card or not.	1,500 (0.69%)	2 values · top: N (90%), Y (10%)
scb_payroll	string	Flag whether the customer used to has Payroll or not.	1,500 (0.69%)	2 values · top: N (80%), Y (20%)
num_children	integer	Total num ber of children.	1,500 (0.69%)	min 0 · median 0 · mean 0 · max 21
age	double	Current Age at the campaign begins.	1,500 (0.69%)	min 23 · median 36 · mean 38 · max 61
income	decimal(12,2)	Income at the campaign begins.	2,401 (1.11%)	min 0 · median 12,253 · mean 16,867 · max 199,626
maxosdc_last_30d	decimal(12,2)	Max Debit Card Outstanding Balance in Last 30 days.	2,401 (1.11%)	min 0 · median 0 · mean 11,355 · max 4,374,289
dcspend_last_30d	decimal(12,2)	Total spending via debit card in last 30 days.	2,401 (1.11%)	min -87,031 · median 0 · mean 165 · max 1,852,988
easypymt_last_30d	decimal(12,2)	Total payment in last 30 days.	2,401 (1.11%)	min 0 · median 373 · mean 1,196 · max 2,000,000

savacc_bal	decimal(12,2)	Total saving account balance in last 30 days.	2,401 (1.11%)	min 0 · median 801 · mean 20,856 · max 4,793,828
currentacc_bal	decimal(12,2)	Total current account balance in last 30 days	2,401 (1.11%)	min 0 · median 0 · mean 0 · max 1
avg_savaccbal_30d	decimal(12,2)	Average of saving account outstanding balance in last 30 days.	2,401 (1.11%)	min 0 · median 993 · mean 15,176 · max 2,471,113
avg_currentaccbal_30d	decimal(12,2)	Average of current account outstanding balance in last 30 days.	2,401 (1.11%)	min 0 · median 0 · mean 30 · max 1,232,520
mob	integer	Month on Bok (unit in number of month). Lower means open any account recently;	2,401 (1.11%)	min 0 · median 40 · mean 55 · max 887
inflow30d	decimal(12,2)	Total inflow in last 30 days. Inflow = cash in to saving or current account.	2,401 (1.11%)	min 0 · median 0 · mean 25,290 · max 21,471,692
outflow30d	decimal(12,2)	Total outflow in last 30 days. Inflow = cash out of saving or current account.	2,401 (1.11%)	min 0 · median 0 · mean 25,546 · max 21,471,691
inflow1_15	decimal(12,2)	Total inflow in last 15 days. Inflow = cash in to saving or current account.	2,401 (1.11%)	min 0 · median 0 · mean 25,290 · max 21,471,692
outflow1_15	decimal(12,2)	Total outflow in last 15 days. Inflow = cash out of saving or current account.	2,401 (1.11%)	min 0 · median 0 · mean 25,546 · max 21,471,691
net_flow_30d	decimal(13,2)	Total Inflow - Outflow in last 30 days	2,401 (1.11%)	min -1,813,780 · median 0 · mean -256 · max 2,295,697
net_flow_15d	decimal(13,2)	Total Inflow - Outflow in last 15 days	2,401 (1.11%)	min -1,813,780 · median 0 · mean -256 · max 2,295,697
label	integer	Campaign Response. 0 = Reject Offer, 1 = Accept Offer in PA Insurance Product, 2 = Accept Offer in Life Insurance Product	–	min 0 · median 0 · mean 0 · max 2

Data quality findings

- **Missing** 1,500 rows have no customer profile at all (every field empty), and only 0.07% of them accepted. They are probably unmatched records, so they are excluded from the analysis. 2,401 rows in total lack financial data.
- **Duplicate** 2,173 rows are exact duplicates. With no customer ID we cannot tell whether they are real repeats, so they are kept.
- **Unusable** The 15-day fields (inflow1_15, outflow1_15, net_flow_15d) are identical to the 30-day fields in 100% of rows. They look like copies, not a separate 15-day window, so they are dropped.
- **Unusable** avg_currentaccbal_30d are zero in more than 99.9% of rows (max currentacc_bal = 1), so they carry no information and are dropped.
- **Suspicious** months on book (mob) = 0 for 56,144 customers (26%), which probably means unknown rather than new. 839 customers have a tenure longer than their adult life (max 887 months).
- **Outlier** num_children has values of 10 or more (3 rows); 17 rows have negative debit-card spend; 342 rows have income = 0. Monetary fields are heavily right-skewed, so medians are used, not means.
- **Encoding** marital_sta is in Thai with 8 overlapping values (e.g. Married / Married registered / Married unregistered), so it is translated and standardised. The data definition notes that main_occupation may be out of date.
- **Context** campaign_month has no year, so seasonality cannot be separated from trend. Monthly volume ranges from 10,179 to 23,625 offers.
- **Imbalance** Only 1.2% of rows are positive, so accuracy is a misleading metric. The model is evaluated with ROC-AUC, PR-AUC and cumulative gains instead.

Acceptance rate = buyers ÷ customers contacted in the group. Analysis excludes the 1,500 rows with no customer profile; categories under 500 customers are folded into "Other"; heatmap and bubble cells need ≥300 customers. Reproduce with `python main.py`.